

Zac Garby

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[GitHub](#)

Experience

University of Nottingham

PhD Student, Functional Programming

2023—present

Nottingham

- Supervised by Professor Graham Hutton in the **Functional Programming Lab**.
- My main research area is **program calculation**: systematically deriving compilers, type systems, and more, from equational specifications. Also interested in **program synthesis** and interactive programming paradigms.

Cogna

Software Engineering Intern, 16 weeks

Summer 2025

London

- Series A startup delivering automated software engineering (i.e. **end-to-end program synthesis**).
- Designed and implemented a tool to resolve ambiguity throughout synthesis, **combining formal methods with LLMs**. Deployed this during my internship, **improving reliability and performance**. Continues to be used and built upon.
- Meaningfully contributed in a small fast-paced team and across the organisation, delivering high quality, well-tested code in Python, Go, and TypeScript with React.

Education

University of Nottingham

MSci, Computer Science

2019—2023

Nottingham

- **First-class Honours** (87% avg., awarded the best overall performance prize)
- Achieved the **highest ever grade** in Computer Science at the University. **Best dissertation** in third year, and **best research project** in my fourth:
Fantasia: Synthesising Recursive Functions Without Trace-complete Examples
Fugue: A Friendly Functional Programming Language with Holes

The Thomas Hardy School

GCSEs and A-levels

2015—2019

Dorchester, Dorset

- A-levels in Mathematics, Further Maths, Computer Science, and Physics (**A*AAA**)

Volunteering

Hackathons UK

2024—2025

Journal of Functional Programming, reviewed several articles 2024—present

President of HackSoc

2021—2023

University of Nottingham Students' Union

- Full responsibility for the society. Measurably **improved engagement**. Delivered talks and workshops. Also **lead organiser** of HackNotts, a 200-person event.

HackNotts '24, sponsored the Haskell prize, taught a workshop 2024

Intl. Conference on Functional Programming, student volunteer 2024, 2025

University of Nottingham, computer science mentor 2020—2021

Thomas Hardy School, founded programming club 2017—2019

Achievements & Recognition

YouTube, 100k+ views of my guest appearances on *Computerphile* 2024, 2025

University of Nottingham, FP lab representative 2023—present

University of Nottingham 2019—2024

- Elizabeth and J D Marsden prize 2023
- Outstanding community contribution prize 2023
- Best overall performance (over the four years) 2023
- High achiever's award (top 5% each year) 2020, 2021, 2022, 2023
- Best dissertation 2022, 2023
- Awarded "silver" scholarship 2019

UKMT, Silver award, senior mathematical challenge 2018

Skills

Haskell, Python, Go, TypeScript, C, $\LaTeX$, Agda, Lean, Rust, HTML, CSS; etc.

Programming language theory/practice, compilers, DSLs, tools. **Full-stack web**, server management, networking. **Media** (graphical and audio, computer vision); electronics & hardware, low-level systems, mixed reality. **AI**, LLM systems programming, security of agentic systems.

Well versed in **mathematics and logic** relating to programming language theory, type theory, semantics & domain theory; category theory, abstract algebra; formal logic & proof mechanisation. Ability to develop formal mathematical theories.

Public speaking, leadership, event planning, logistics, finance. **Educating**, and engaging a range of audiences. Qualified first aider.

Publications

Calculating Compilers Effectively

Zac Garby, Graham Hutton, and Patrick Bahr
Haskell Symposium 2024

The Calculated Typer

Zac Garby, Patrick Bahr, and Graham Hutton
Haskell Symposium 2025

The LLMbda Calculus

Zac Garby, Andrew D. Gordon, David Sands
Pending publication

Sound-By-Construction Type Systems

Patrick Bahr, Zac Garby, Graham Hutton
Pending publication

Conferences

- **Haskell Symposium** 2024, 2025
- **ICFP** 2024, 2025
- **Midlands Graduate School** 2022, 24, 26

Hackathons

- **NIO Hackathon**, 2026 **1st place**
- **Royal Hackaway v8**, 2025 **2nd place**
- **HackNotts '24**, 2024 **special prize**
- **AstonHack 9**, 2024 **2nd place**
- **HackSussex**, 2024 **1st place**
- **Royal Hackaway v7**, 2024 **1st place**
- **HackSussex**, 2023 **1st place**
- **HackNotts**, 2023 **lead organiser**
- **OxfordHack**, 2022 **sponsor prize**
- **HackNotts**, 2022 **an organiser**
- **AstonHack**, 2021 **1st place** + other prize
- **HackNotts**, 2020 **sponsor prize**
- And more. View my [portfolio](#).